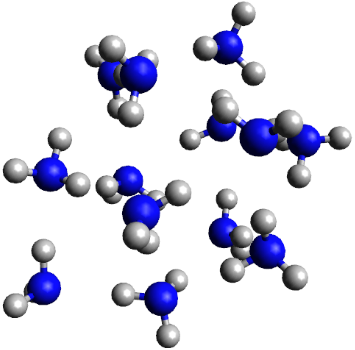
# Thermodynamically Consistent Coarse-Graining of Molecular Systems

## Beyond Rigid-Monomer Models

João Pimentel

Mandelshtam Group
Department of Chemistry
University of California, Irvine

# Coarse-Graining



3 coordinates per atom

$\Downarrow$

6 coordinates per molecule

Treating groups/molecules as rigid bodies (CG-fixed) is too coarse.

# Coarse-Grained Free Energy

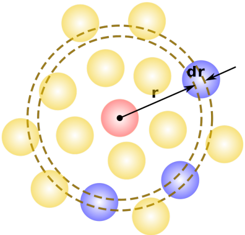
- Boltzmann distribution:

$$P(\mathbf{r};T) = Z(T)^{-1} e^{-\beta V(\mathbf{r})}, \qquad Z(T) = \int d\mathbf{r}\, e^{-\beta V(\mathbf{r})}$$

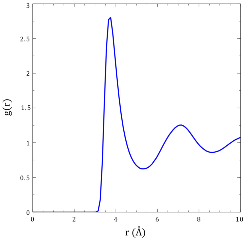
- Coarse graining $\mathbf{r} \mapsto (\mathbf{R}_{\mathsf{CG}}, \mathbf{q})$:

$$Z(T) = \int d\mathbf{R}_{\mathsf{CG}}\, e^{-\beta F(\mathbf{R}_{\mathsf{CG}};T)}, \qquad e^{-\beta F(\mathbf{R}_{\mathsf{CG}};T)} \propto \int d\mathbf{q}\, e^{-\beta V(\mathbf{R}_{\mathsf{CG}},\mathbf{q})}$$

# Can we compute the integral?




- Bottom–up CG methods avoid full FES by matching observables.



# Our Approach

Exploit near-harmonic nature of fast modes.

- Relax fast modes (adiabatic approximation):

$$V_0(\mathbf{R}) := \min_{\mathbf{q}} V(\mathbf{R}, \mathbf{q})$$

- Compute vibrational frequencies from the Hessian at the minimum:

$$\mathbf{M}^{-1/2} \nabla \nabla^\top V \, \mathbf{M}^{-1/2}$$

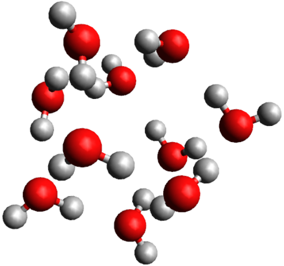
- Integrate out fast modes using the harmonic approximation:

$$F(\mathbf{R}; T) \approx V_0(\mathbf{R}) + \begin{cases} \sum_k \left[ \frac{\hbar\omega_k}{2} + \frac{1}{\beta} \log\left(1 - e^{-\beta\hbar\omega_k}\right) \right] & \text{(Quantum)} \\ \frac{1}{\beta} \sum_k \log(\beta\hbar\omega_k) & \text{(Classical)} \end{cases}$$

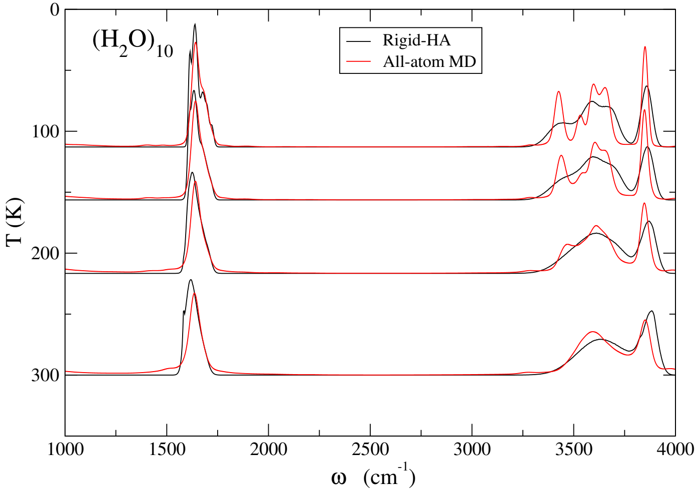
- We sample this FES directly using Monte Carlo.
- May include NQEs! No need for PIMD.

# Proof of Concept and Simulation Setup

- Method validated (benchmarked against classical AA simulation) for:
  - $(NH_3)_{13}$ (CHARMM-like PES)
  - $(H_2O)_6$ and $(H_2O)_{10}$ (q-TIP4P/F)
- Numerical setup:
  - Replica Exchange Monte Carlo (REMC) with $J = 20$ replicas
  - Temperature range: $20$–$300$ K, geometric schedule



# Results: Spectrum of Water Decamer




Main features such as the OH-stretch (broader due to strong hydrogen-bonding interactions) and HOH-bend (narrower) are preserved.

# Calculating Energy and Heat Capacity

$i$ = molecule index, $N_i$ = atoms per molecule,
$I$ = number of molecules, $N = \sum_i N_i$ = total number of atoms

- Canonical average energy:

$$U(T) = 3Ik_{\mathrm{B}}T + \langle E(\mathbf{R}_{\mathsf{CG}};T)\rangle_T$$

where

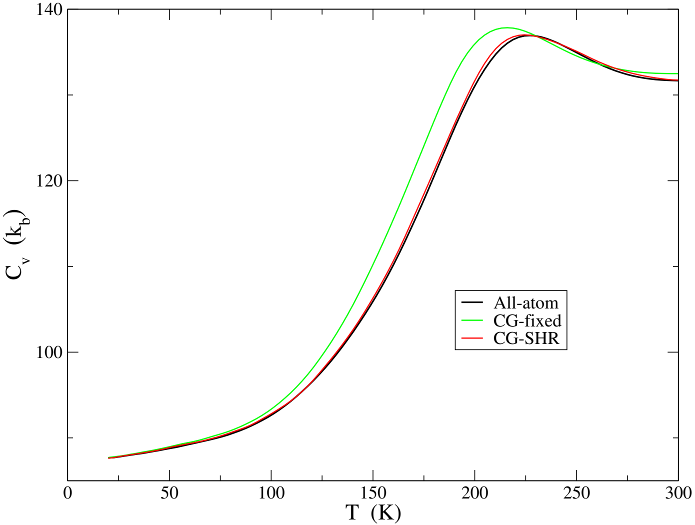
$$E(\mathbf{R}_{\mathsf{CG}};T) = \frac{\partial[\beta F(\mathbf{R}_{\mathsf{CG}};T)]}{\partial\beta} = V_0(\mathbf{R}_{\mathsf{CG}}) + \sum_{k=1}^{L}\left(\frac{\hbar\omega_k}{2} + \underbrace{\frac{\hbar\omega_k}{e^{\beta\hbar\omega_k}-1}}_{\text{depends on } T}\right)$$

- Heat capacity:

$$C_v(T) = 3Ik_{\mathrm{B}} + \frac{1}{k_{\mathrm{B}}T^2}\left[\langle E^2(\mathbf{R}_{\mathsf{CG}};T)\rangle_T - \langle E(\mathbf{R}_{\mathsf{CG}};T)\rangle_T^2\right] - \frac{1}{k_{\mathrm{B}}T^2}\left\langle\frac{\partial E(\mathbf{R}_{\mathsf{CG}};T)}{\partial\beta}\right\rangle_T$$

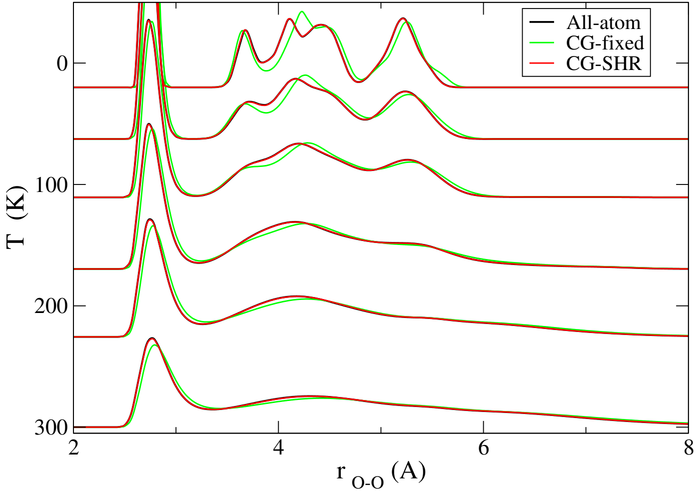
$$\frac{\partial E(\mathbf{R}_{\mathsf{CG}};T)}{\partial\beta} = -\sum_{k=1}^{L}\frac{\hbar^2\omega_k^2 e^{\beta\hbar\omega_k}}{(e^{\beta\hbar\omega_k}-1)^2}$$

# Results: Heat Capacity of Water Decamer




SHR agrees with all-atom calculation. CG-fixed deviates.

# Results: Oxygen-Oxygen Radial Distribution Function




Curves shifted vertically by $T$.

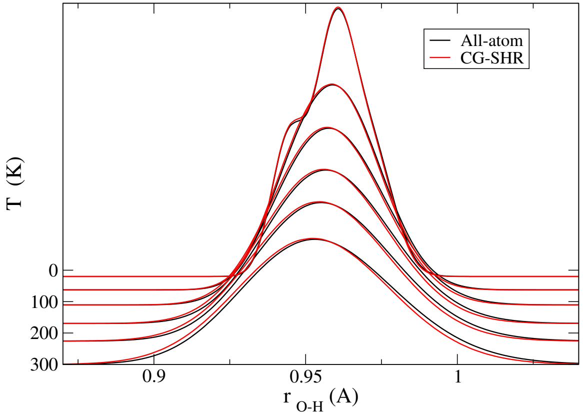
# Recovering Atomistic Detail

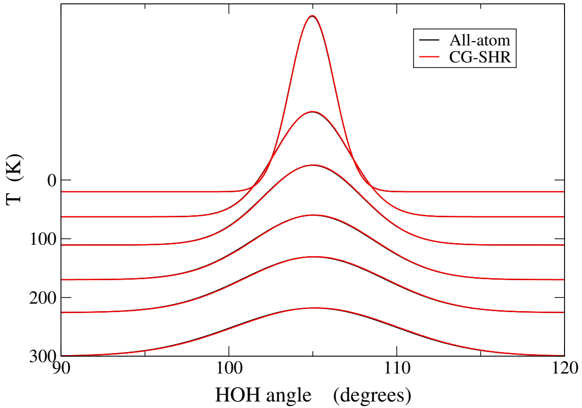
Add harmonic fluctuations along the fast normal modes.

$$\mathbf{r} = \mathbf{r}^* + \sum_{l=1}^{L} \xi_l \, \mathbf{U}_l, \qquad \xi_l \sim \mathcal{N}\left(0, (\sigma_l)^2\right)$$

$$(\sigma_l)^2 = \begin{cases} \dfrac{\hbar}{2\omega_l} \coth\left(\dfrac{\beta\hbar\omega_l}{2}\right) & \text{(Quantum)} \\ \dfrac{1}{\beta(\omega_l)^2} & \text{(Classical)} \end{cases}$$

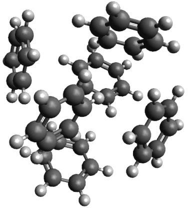
# Results: Bond Lengths and Angles

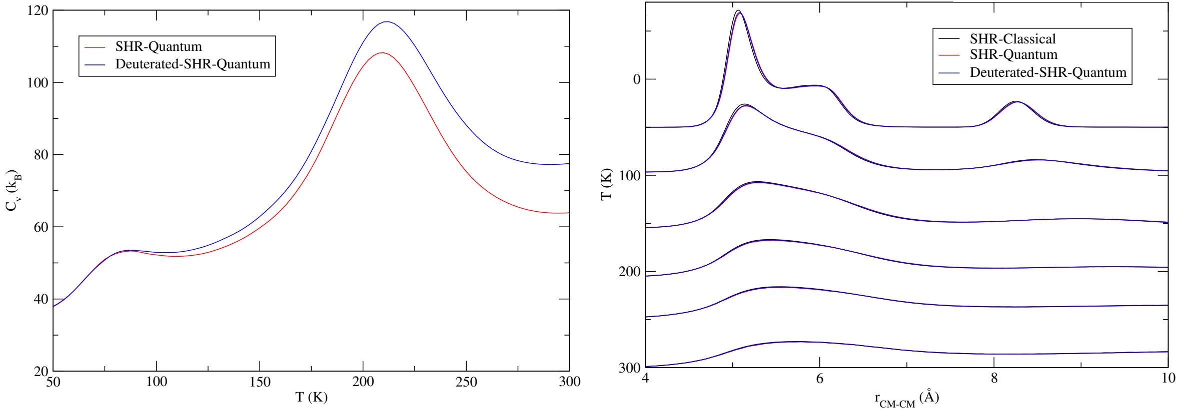



O-H bond lengths and H-O-H angles reconstructed accurately. Curves shifted vertically by $T$.

# NQEs at no extra cost: Benzene Hexamer





# Key Takeaways

$i$ = molecule index, $N_i$ = atoms per molecule,
$I$ = number of molecules, $N = \sum_i N_i$ = total number of atoms

- MC integration in low-dimensional subspace ($6I$ instead of $3N$) avoids the curse of dimensionality.
- Requires fewer Monte Carlo steps to converge.
- Block-diagonal Hessian + Newton-Raphson maintain accuracy with minimal cost ($\sim \sum_i (3N_i)^3$ vs. $\sim (\sum_i 3N_i)^3$ for diagonalization).
- Enables approximate all-atom reconstruction.

# tl;dr

$F(\mathbf{R}_{\mathsf{CG}}; T)$ by integrating out $\mathbf{q}$ with the harmonic approximation

CG efficiency with (near) all-atom thermodynamic consistency

# Next Steps

- **Bridge ab initio and empirical PES:**
  - Ab initio (bottom up): clean electronic PES, no NQEs.
  - Empirical (top down): NQEs are implicit; transfer across isotopes is poor.
  - We can start from ab initio, add NQEs in fast modes, get an effective classical $F(\mathbf{R}_{\mathsf{CG}}; T)$ with isotope control.
- **Accelerate with ML:** ML-based parameterization of $F(\mathbf{R}_{\mathsf{CG}}; T)$.

# References

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# Acknowledgments